

1855. Maximum Distance Between a Pair of Values

Solved 

Medium  Topics  Companies  Hint

You are given two **non-increasing 0-indexed** integer arrays `nums1` and `nums2`.

A pair of indices (i, j) , where $0 \leq i < \text{nums1.length}$ and $0 \leq j < \text{nums2.length}$, is **valid** if both $i \leq j$ and $\text{nums1}[i] \leq \text{nums2}[j]$. The **distance** of the pair is $j - i$.

Return the **maximum distance** of any **valid pair** (i, j) . If there are no valid pairs, return `0`.

An array `arr` is **non-increasing** if $\text{arr}[i-1] \geq \text{arr}[i]$ for every $1 \leq i < \text{arr.length}$.

Example 1:

Input: `nums1 = [55,30,5,4,2]`, `nums2 = [100,20,10,10,5]`

Output: 2

Explanation: The valid pairs are $(0,0)$, $(2,2)$, $(2,3)$, $(2,4)$, $(3,3)$, $(3,4)$, and $(4,4)$.

The maximum distance is 2 with pair $(2,4)$.

Example 2:

Input: `nums1 = [2,2,2]`, `nums2 = [10,10,1]`

Output: 1

Explanation: The valid pairs are $(0,0)$, $(0,1)$, and $(1,1)$.

The maximum distance is 1 with pair $(0,1)$.

Example 3:

Input: `nums1 = [30,29,19,5]`, `nums2 = [25,25,25,25,25]`

Output: 2

Explanation: The valid pairs are $(2,2)$, $(2,3)$, $(2,4)$, $(3,3)$, and $(3,4)$.

The maximum distance is 2 with pair $(2,4)$.

Constraints:

- $1 \leq \text{nums1.length}, \text{nums2.length} \leq 10^5$
- $1 \leq \text{nums1}[i], \text{nums2}[j] \leq 10^5$
- Both `nums1` and `nums2` are **non-increasing**.

Python:

```
class Solution:
    def maxDistance(self, A: List[int], B: List[int]) -> int:
        i, j = 0, 1

        while i < len(A) and j < len(B):
            i += A[i] > B[j]
            j += 1

        return j - i - 1
```

JavaScript:

```
const maxDistance = (A, B) => {
    let i, j;

    for (i = 0, j = 1; i < A.length && j < B.length; j++)
        i += A[i] > B[j];

    return j - i - 1;
};
```

Java:

```
class Solution {
    public int maxDistance(int[] A, int[] B) {
        int i, j = 1;

        for (i = 0; i < A.length && j < B.length; j++)
            if (A[i] > B[j])
                i++;

        return j - i - 1;
    }
}
```